

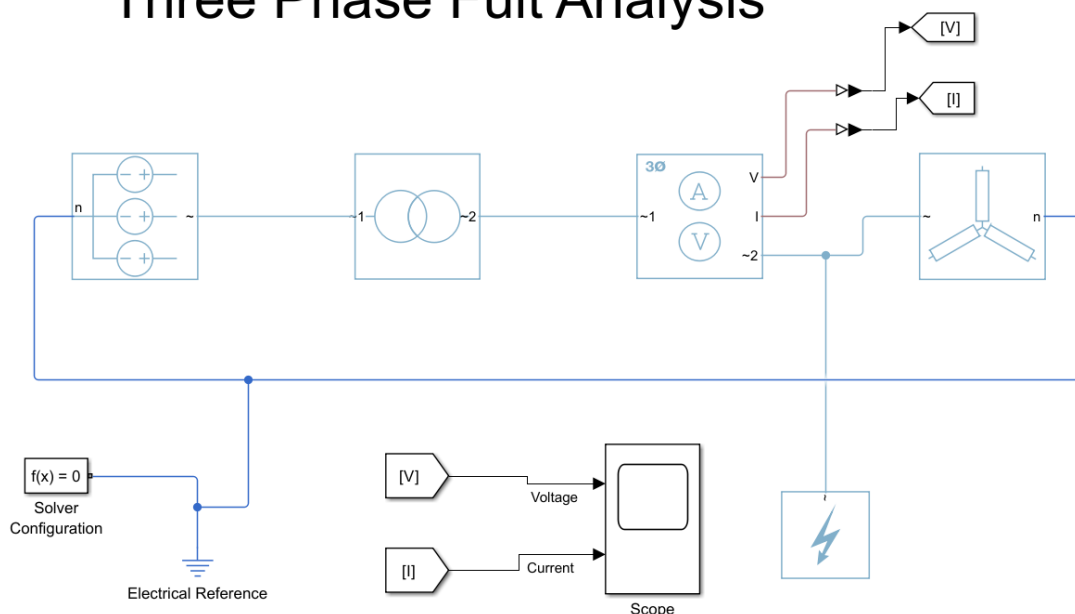
Three-Phase Fault Analysis – Project Description

This project presents a simple **three-phase fault analysis** model built in **Simulink** and **Simscape Electrical**. The objective is to analyze the behavior of a balanced three-phase system when a fault is applied and to observe the corresponding changes in voltage and current.

The model consists of a three-phase AC voltage source feeding a transmission line and a balanced load. A configurable fault block is inserted on the line to simulate short-circuit conditions. Measurement blocks are used to obtain line voltages and currents, which are then displayed on a scope and exported for plotting.

All signals are referenced to a common electrical ground, and proper solver configuration ensures numerical stability during the transient fault event. The simulation allows the user to visualize how voltages drop and currents rise sharply during the fault, demonstrating fundamental power system behavior.

Three Phase Fult Analysis



This project includes:

- The **Simulink schematic** of the three-phase system
- A **plot of measured voltage and current** during the fault
- A lightweight and beginner-friendly setup suitable for electrical engineering students

The repository provides the model file and documentation to help others explore basic power system fault analysis using Simscape Electrical.

A plot of measured voltage and current during the fault

